Tushar Bansal

+918209100349 tusharbansal030@gmail.com /in/tusharr-bansal /Tusharr-Bansal portfolio

Education

Jaipur Engineering College and Research Centre **Sep. 2023 – May 2027**
B.Tech. (IT); ***CGPA: 9.16*** *Jaipur, Rajasthan*

Indian Institute of Technology, Madras **Sep. 2021 – Apr. 2027**
B.S. (Data Science and Applications); ***CGPA: 7.68*** *Distance Learning*

Technical Skills

Languages: Python, SQL, Java, C++, R, Bash
ML & Deep Learning: PyTorch, Hugging Face, Scikit-Learn, LightGBM, NumPy, Pandas, Matplotlib
LLMs & Agents: LangGraph, LangChain, RAG (hybrid BM25 + vector search), Gemini API, Prompt Engineering
ML Evaluation: Precision/Recall@k, mAP, nDCG, MRR, F1, bootstrap confidence intervals, LLM-as-judge
MLOps: ONNX export & int8 quantization, Langfuse (tracing, prompt versioning), eval pipelines, Docker, CI/CD, Render
Backend & Data: FastAPI, Flask, REST APIs, PostgreSQL, pgvector, Redis, Celery, Git, Linux

Experience

Molde University College, Norway **May 2026 – Present**
Research Intern (Supervisor: Dr. Swati Aggarwal, Professor of AI) *Molde, Norway (Remote)*

- Designing and building a **trustworthy pediatric heart-screening** system that runs **offline** on a smartphone, able to **detect murmurs and abnormality**, combining signal-quality checks, **self-supervised learning**, **uncertainty-based rejection** and **explainable AI** on **3,163 CirCor recordings (942 patients)**, hand-labeled for signal quality.

Creware Technologies Pvt. Ltd. **June 2025 – August 2025**
Software Engineer Intern *Jaipur, Rajasthan (On-Site)*

- Architected a Rasa-based conversational **AI assistant using Rasa NLU and SDK**, optimizing intent classification across **1,000+ training variants per intent** and designing multi-turn booking workflows that validated user inputs, **automated backend API transactions**, and triggered operational email notifications.

Indian Institute of Technology, Madras **May 2023 – May 2024**
Teaching Assistant for DBMS *Chennai, Tamil Nadu (On-Site)*

- Supported instruction for **300+ students in IIT Madras' DBMS course**, guiding learners through database design (RDBMS, normalization), indexing (Hash tables, B/B+ Trees), and transaction management (ACID compliance), while building practical PostgreSQL exercises for hands-on learning.

Publications & Research Papers

T. Bansal et al., **"Clustering Districts by Educational Performance Using Machine Learning and Neutrosophic Logic"***, Presented at Springer International Conference on Emerging Trends in Engineering and Smart Systems (ICETESS), Jaipur, India, 2026.* | *Presented Paper*

Projects

FindXRay: X-ray Case Search Agent | *LangGraph, Gemini, ONNX, Langfuse* | *Live* | *GitHub* **October 2026**

- Raised report agreement with radiologists by **33%** (**CheXbert F1 0.50 vs 0.37**, 95% CI) and found matching cases at **1.7× chance** (**precision@5 0.42**, 720 held-out X-rays) by building a LangGraph agent that drafts source-cited reports from similar-case retrieval, **int8-quantized** to run in-browser in 0.5 s.

SeeCite: Research-Paper RAG Agent | *pgvector, BM25, Gemini, FastAPI* | *Live* | *GitHub* **October 2026**

- Delivered **91% answer correctness** with **zero fabricated claims** (blind 60-answer review) and **recall@10 of 0.78** vs 0.64 for vector-only search by building a hybrid **BM25 + pgvector** RAG agent with code-validated citations over 29 papers, defaulting to single search after evals showed it matched the agent at **3**× fewer LLM calls.

Music Genre Classification under Distribution Shift | *PyTorch, Transformers* | *Link* **April 2026**

- Mitigated distribution shift in music genre classification by engineering a data augmentation pipeline with **5,000+ synthetic audio mashups** and ESC-50 environmental noise, then benchmarking CNN, CNN-BiLSTM, and **86M-parameter Audio Spectrogram Transformer** models to achieve a peak **0.89555 F1-score** on Kaggle using mixed-precision training, test-time augmentation, and label smoothing.

Certifications

Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate | *Certificate Link* **October 2025**
Oracle Cloud Infrastructure 2025 Certified Generative AI Professional | *Certificate Link* **October 2025**
Oracle AI Vector Search Certified Professional | *Certificate Link* **October 2025**